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Data-driven state of health estimation for lithium-ion battery based on voltage variation curves

Jiang Wu, Zelong Liu, Yan Zhang, Dong Lei, Bo Zhang, Wen Cao *

School of Electronics and Information, Xi'an Polytechnic University, Xi'an 710048, China
Xi'an Key Laboratory of Interconnected Sensing and Intelligent Diagnosis for Electrical Equipment, Xi'an 710048, China

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ABSTRACT

The state of health (SOH) estimation of lithium-ion batteries (LIBs) is crucial for battery management system, but the accuracy and generalizability of the widely used data-driven methods are strongly dependent on the selection of LIBs health features (HFs). In this paper, four types of LIBs with different anode types from four datasets, including NASA dataset, CALCE dataset, Oxford dataset and UL-PUR dataset, were selected to extract the area of constant current charging and discharging voltage curves as two sets of HFs, and then the high correlation between the HFs and SOH is verified by their Pearson coefficient. Secondly, with the two sets of HFs, the SOH of selected batteries in the four datasets are evaluated under Gaussian Process Regression, Long and Short-Term Memory neural network and Back Propagation neural network respectively. With a training/test set ratio model of 50/50 and cross-validation method, all algorithms obtain accurate SOH estimation results. Finally, the estimation results are compared with reference data under the same dataset and training mode, and it is found that the proposed method shows better estimation accuracy and robustness than other evaluation methods by multiple HFs or even complex algorithms.

1. Introduction

Due to the long lifetime, high energy density and small size, lithium-ion batteries (LIBs) are widely used in electric vehicles (EVs) [1,2]. When LIBs are used as power supply, an accurate online assessment of operating status is important for the battery management system (BMS), which determines the service life and even the safety of the EV [3]. Therefore, the assessment for the state of health (SOH), which is considered one of the important indicators of battery performance, is proposed by researchers and institutes [4–6]. The SOH is defined as the ratio of the current maximum capacity of the battery to its initial rated capacity [7], and the SOH generally decreases with the aging circle of the battery charge and discharge, which is closely related to their internal electrochemical reactions [8]. When the SOH falls below 80 %, the LIBs are usually considered to reach the end-of-life and no longer meet the power and energy requirements of the devices [9].

Currently, the estimation methods of SOH are mainly divided into types of model-based and data-driven [10]. Among the model-based methods, the electrochemical and equivalent circuit models are often used. However, the presence of partial differential equations in the electrochemical model leads to complex calculations, which weakens the effectiveness of real-time estimation [11]. Moreover, due to the fixed structure of the equivalent circuit model, high-accuracy SOH estimation for the whole life cycle of batteries are also difficult to achieve [12].

In recent years, substantial development of machine learning techniques enables data-driven algorithms to be widely used in the SOH estimation of LIBs [13]. Gong proposed an automatic encoding-decoding model based on deep learning to establish the mapping between charging curves properties and SOH of the battery [14]. Shi proposed a fast SOH estimation method using a single linear short-term feature from the charging curve property [15]. Chang considered two definitions of SOH based on battery cell aging and consistent aging and only a consistent model based on Copula theory and prediction of the battery parameters using Gaussian process regression (GPR) should be carried out without any training data [16]. Deng proposed the use of early battery data to achieve pattern recognition and transfer learning, both of which can effectively improve the accuracy of SOH estimation, and extract HFs and input to LSTM to build a SOH estimation model [17]. Qian used the historical state information and load information of lithium batteries to build an attention-based multisource [18]. He

* Corresponding author at: School of Electronics and Information, Xi'an Polytechnic University, Xi'an 710048, China.
E-mail address: caowen@xpu.edu.cn (W. Cao).

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proposed a new method for estimating the health state (SOH) and remaining useful life (RUL) of LIBs using Dempster-Shafer theory and Bayesian Monte Carlo methods, a new method is developed to establish an empirical model based on the physical degradation behavior of lithium-ion batteries [19]. Fan et al. proposed a hybrid neural network based on a Gate Recursive Unit-Convolutional Neural Network, which can utilize a deep learning technique to learn the shared information and time dependencies of the charging curves [20]. In a word, there are various data-driven algorithms for SOH estimation recently, but it is necessary to feed the algorithm with features that adequately reflect the SOH of the LIBs.

The health features (HFs) extracted from the operating data of the battery are frequently selected as the input to the algorithm. Guo utilized correlation analysis and principal component analysis to extract and optimize 18 HFs extracted from the charging stage and developed a SOH estimation model using a relevance vector machine [21]. Wang presented a SOH estimation based on small sample data, in which twelve HFs were extracted from incremental capacity curves, and finally obtained a structurally weighted dual support vector regression algorithm (SVR) to estimate the SOH [22]. Son proposed a fusion framework for SOH estimation using multivariate physical features, in which nine HFs such as mechanical evolution features and impedance correlation were extracted, and the SOH of LIBs was estimated by GPR [23]. Jia extracted eight indirect HFs from the voltage, current, and temperature curves of the battery under charge and discharge, and finally used GPR to estimate the SOH and remaining life of LIBs [24]. Zhao extracted six HFs from the charging voltage curve to Long and short-term memory (LSTM) and GPR to finally obtain the SOH and remaining life of the LIBs [25]. Feng extracted six HFs from the charging current curve of the battery and then used an improved GPR to estimate the SOH and remaining life [26]. Guo preprocessed the voltage data during constant current (CC) charging of batteries to suppress the measurement noise and facilitate the calculation of the differences in voltage profiles for different aging levels, and then extracted six HFs from them and used an ensemble SVM model to establish the relationship between HFs and battery SOH [27]. Goh et al. also proposed a SOH estimation by discrete curvature features in discharge curve of LIBs [28]. Guo considered the capacity recovery phenomenon of LIBs and used wavelet analysis to decompose the extracted six HFs, and the SOH of the battery was estimated by a deep bidirectional LSTM [29]. Driscoll proposed three HFs extracted from the voltage and current temperature profiles of the battery and used artificial neural networks to build a SOH estimation model [30]. In summary, existing HF extraction methods often rely on least squares identification of battery model parameters or utilize a smoothing curve approach to extract distinctive features. However, current SOH estimation methods all have the following challenges:

(1) The extraction of a large number of HFs leads to computational resource wastage and potential model overfitting.
(2) Existing methods primarily focus on lithium batteries with a single anode material type, neglecting the exploration of estimation methods for batteries with different anode materials.
(3) The increasing complexity of improved data-driven algorithms overlooks the computing power limitations of BMS.

To address the computational conditions of deep learning and complex HFs extraction, we selected the two areas of the voltage variation curve vs. time under CC charge and discharge as the two HFs to estimate the SOH of LIBs. Then, the Pearson's coefficient between the extracted HFs and SOH value was calculated and analyzed. To verify the effectiveness of the proposed method on different types of LIBs, the NASA battery dataset, the CALCE battery dataset, the UL-PUR battery dataset, and the Oxford battery dataset were selected for validation respectively. Finally, the basic GPR, LSTM, and Back propagation (BP) are chosen to estimate the SOH of LIBs, and the estimation results were comprehensively compared with references.

Therefore, this paper makes the following main contributions:

(1) We propose an approach that extracts only two HFs with a high correlation to SOH, thereby optimizing computational resources and reducing the risk of overfitting.
(2) Our method is validated using datasets from four batteries with diverse anode material types, demonstrating the generalization ability of the estimation method across different types of batteries.
(3) By employing three commonly used algorithms with two HFs, we achieve high accuracy in SOH estimation compared to existing literature, while considering the computational limitations of BMS.

2. Battery dataset

Four battery datasets were chosen as the experimental sample, containing the NASA Battery Dataset [31], the CALCE Battery Dataset [19], the Oxford Battery Dataset [32], and the Underwriters Laboratories Inc.-Purdue University (UL-PUR) Battery Dataset [33].

2.1. NASA dataset

The NASA battery dataset is from the NASA Ames Prediction Centre of Excellence, and the aging data of four LIBs, serial numbers B0005, B0006, B0007, and B0018, were used in this paper. The batteries were made of $LiCoO_2$/$LiNiCoAlO_2$ as the cathode material and graphite as the cathode material. Under 24 °C, these four batteries were charged at 1.5A-CC until the battery voltage reached 4.2 V, and then charged at a constant voltage (CV) of 4.2 V until the charge current was less than 20 mA. During discharging, all the batteries were discharged at 2A-CC until the voltage of batteries B0005, B0006, B0007, and B0018 dropped to 2.7 V, 2.5 V, 2.2 V, and 2.5 V respectively. The degradation data for the four batteries are shown in Fig. 1(a).

2.2. CALCE dataset

The CALCE battery dataset is opened by the University of Maryland. In this study, INR18650 prismatic batteries, numbered CS_2_35, CS_2_36, and CS_2_37, were used, while the positive and negative electrode material of the batteries was $LiCoO_2$ and graphite. The batteries were charged at 0.5C-CC until the voltage reached 4.2 V, and then charged at 4.2V-CV until the charge current was less than 50 mA. On the other hand, the three batteries were discharged at 1C-CC until the 2.7 V discharge cut-off voltage. The degradation data for the three batteries are shown in Fig. 1(b).

2.3. UL-PUR dataset

The battery dataset selected from the UL-PUR open-source dataset is a ternary LIBs 18650 with $LiNiCoAlO_2$ and graphite as the positive electrode. The battery was charged and discharged at 0.5C-CC at room temperature with a charge cut-off voltage and discharge cut-off voltage of 4.2 V and 2.7 V, respectively. The degradation data for the two chosen batteries with number #1 and #2, are shown in Fig. 1(c).

2.4. Oxford dataset

The Oxford Battery Dataset is a publicly available battery aging dataset from the University of Oxford, consisting of degradation data from 8 pouch LIBs manufactured by Kokammad. The eight batteries, numbered from 1_1 to 1_8, have a nominal capacity of 740mAh and a nominal voltage of 4.2 V under 40 °C with $LiCoO_2$/$LiNiMnCoO_2$ as the cathode material and graphite as the cathode material. For degradation, the batteries were charged under CC and discharged in the condition of Artemis city driving cycles. Meanwhile, the batteries were charged and

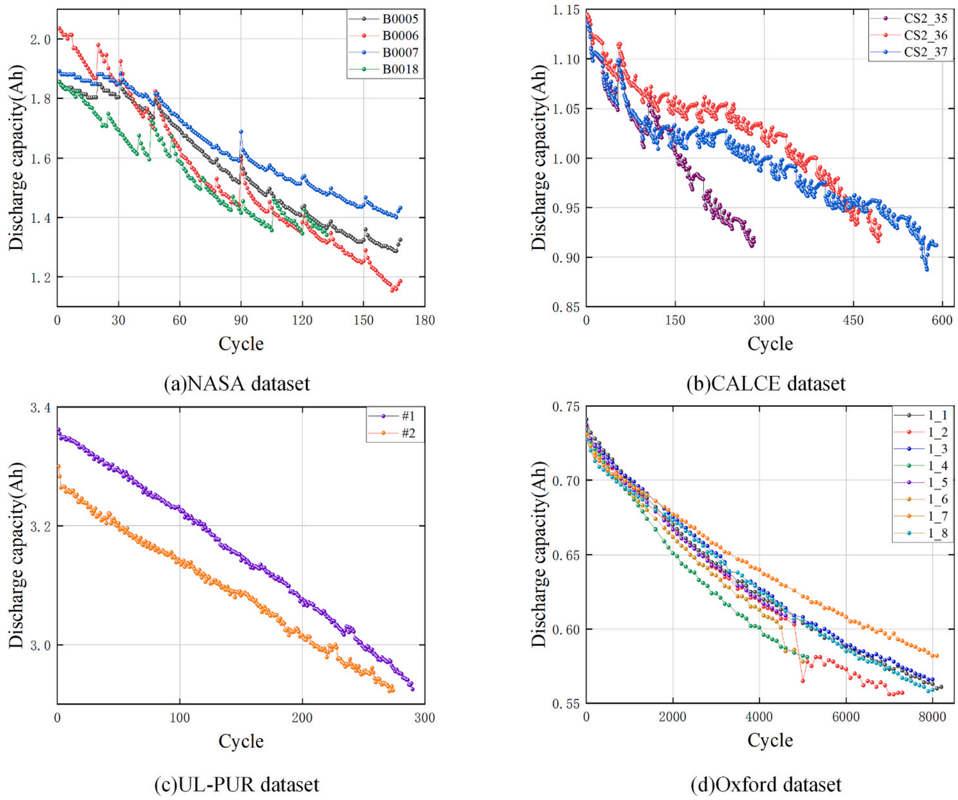


Fig. 1. Discharge capacity of the different dataset batteries.

discharged using 1C-CC every 100 degradation cycles with a discharge cut-off voltage of 2.7 V. The degradation curves for the eight batteries are shown in Fig. 1(d).

3. Feature extraction

To accurately estimate SOH, it is necessary to input appropriate HFs which can fully reflect the SOH variation of LIBs to the algorithm. The simplicity and accuracy of the selected HFs can easily affect the reliability and effectiveness of the SOH estimation. Actually, in the charging and discharging process, the time integral of the voltage curve multiplied by the current can obtain the charging and discharging energy of batteries, and the energy is directly connected with the discharging capacity of batteries, so the variation of the area of the CC charging and discharging voltage curve can very well map the capacity degradation process of batteries.

For the CV charging process, the integration of the current variation curve vs. time of the battery was proposed by Feng and Shi as the HFs [26]. However, due to the short duration of the CV charging and the inadequate sampling points, the area calculation of the current curve has a large error, which makes it is not suitable for practical applications. Therefore, we directly extracted the area of the voltage variation curve vs. time during CC charging and CC discharging as the HFs, which can reliably reflect the real SOH of LIBs.

3.1. Feature extraction based on CC charging curves

Taking the typical battery from the four datasets, the voltage variation curve was analyzed. The terminal voltage varying with time of B0006 battery of the NASA dataset, CS_2_35 battery of CALCE dataset, the #1 battery of UL-PUR dataset and 1_1 battery of Oxford dataset is shown in Fig. 2(a) to (d), respectively.

Fig. 2(a)–(d) shows that as the number of aging cycle increases, the voltage curve of the battery gradually moves to the left, which represents the gradually increasing rate of voltage variation and less time to reach the cut-off voltage. Therefore, the area formed by the voltage curve and the *x*-axis, i.e. time, becomes increasingly smaller with the degradation degree of the battery.

The formula for calculating the area of the voltage curve under CC charging is shown in Eq. (1).

$$S_{ch} = \int_{t_1}^{t_2} V_{ch}(t)dt \tag{1}$$

where t_1 and t_2 represent the start time and end time of charging respectively, and $V_{ch}(t)$ is the voltage over time of the CC charging.

3.2. Feature extraction based on CC discharging curves

After fully charged, the battery is discharged at CC mode. Similarly, only the battery terminal voltage vs. time during discharging is needed for feature extraction. Due to the battery degradation, the total discharging time becomes increasingly shorter, and the terminal voltage decreases at a faster rate. However, the drop rate of the terminal voltage requires extensive calculations, which is not suitable for online application. Thus, we directly extracted the area of the voltage curve vs. time during CC discharging as a HF.

As an example, the terminal voltage of the B0006 battery of the NASA dataset, CS_2_35 battery of CALCE dataset, 1_1 battery of Oxford dataset and #1 battery of UL-PUR dataset is shown in Fig. 3(a) to (d), respectively.

The formula for calculating the area of the discharging voltage curve under CC is shown in Eq. (2).

$$S_{dis} = \int_{t_3}^{t_4} V_{dis}(t)dt \tag{2}$$

where t_3 and t_4 represent the start time and end time of discharge respectively, and $V_{dis}(t)$ is the voltage over time during the CC

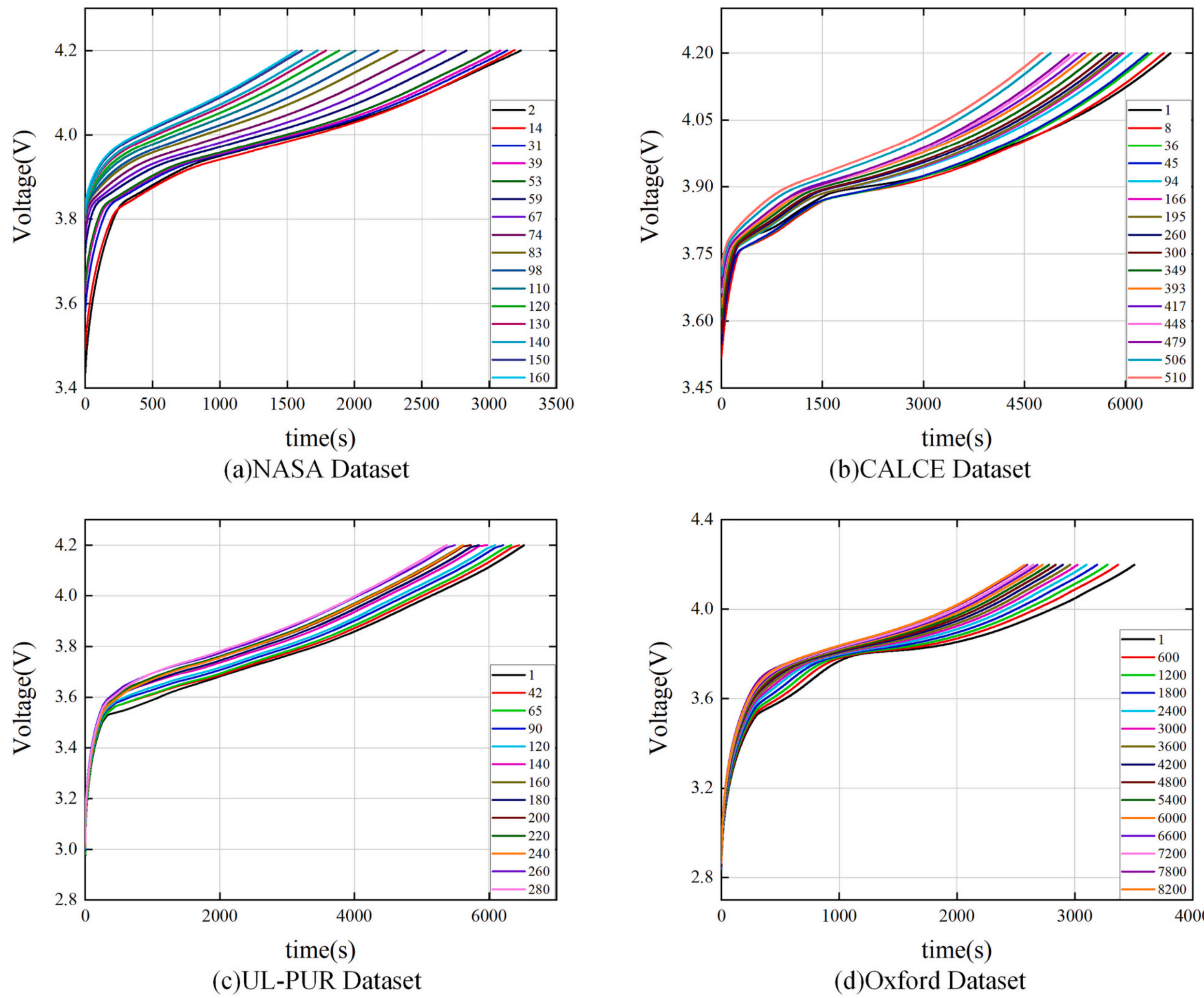


Fig. 2. The variation of voltage profiles of different dataset in CC charging with aging cycles.

discharging.

3.3. Correlation analysis

The Pearson correlation coefficient, denoted as *r,* is one of the conventional correlation coefficients to measure the degree of linear correlation between two variables, such as X and Y. The value of r ranges from −1 to 1, with higher absolute values indicating a stronger correlation. The formula for calculating the correlation coefficient is shown in Eq. (3).

$$r = \frac{\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \overline{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \overline{Y})^2}} \tag{3}$$

where $\overline{X}$ and $\overline{Y}$ are the mean values of X and Y respectively, and n is the number of samples.

Table 1 shows the correlation coefficients between the two extracted HFs in charging/discharging voltage curve and their SOH as S_{ch}-SOH and S_{dis}-SOH. As shown, the S_{ch}-SOH and S_{dis}-SOH of the batteries in the NASA dataset and the batteries in the CALCE dataset are almost all greater than 0.9, the two correlation coefficients of the eight batteries in the Oxford dataset are almost all greater than 0.99, the S_{ch} -SOH of both batteries in the UL-PUR dataset are greater than 0.9, and the S_{dis}-SOH is about 0.83. The above correlations between HFs and battery discharge capacity are all above 0.8, indicating that the calculation of the area of the voltage curve during battery charging and discharging can well substitute the battery charging and discharging energy, and can explain the battery capacity decline process well, indicating that the changes of these two HFs can map the SOH change process properly.

4. Methodology

In this study, the variation of voltage curves during charging and discharging of LIBs is proposed to estimate the SOH of the battery. Fig. 4 depicts the experimental validation framework of the proposed SOH estimation model. By selecting the appropriate four battery datasets from the official dataset website and choosing to extract the area of the battery charge and discharge voltage curve as HFs. Two testing modes were chosen: in Mode 1 the first 50 % of the dataset was used as the training set and the second 50 % as the test set, and Mode 2 adopts the cross-validation method, i.e., the data of one battery is used as the test set and the data of the remaining batteries are used as the training set. Moreover, three algorithms, including GPR, LSTM and BP, were used to validate the effectiveness of our proposed HFs for SOH estimation for different anode types of batteries. The remaining procedures are all described below.

4.1. Gaussian process regression

GPR is a non-parametric model for data regression analysis by a

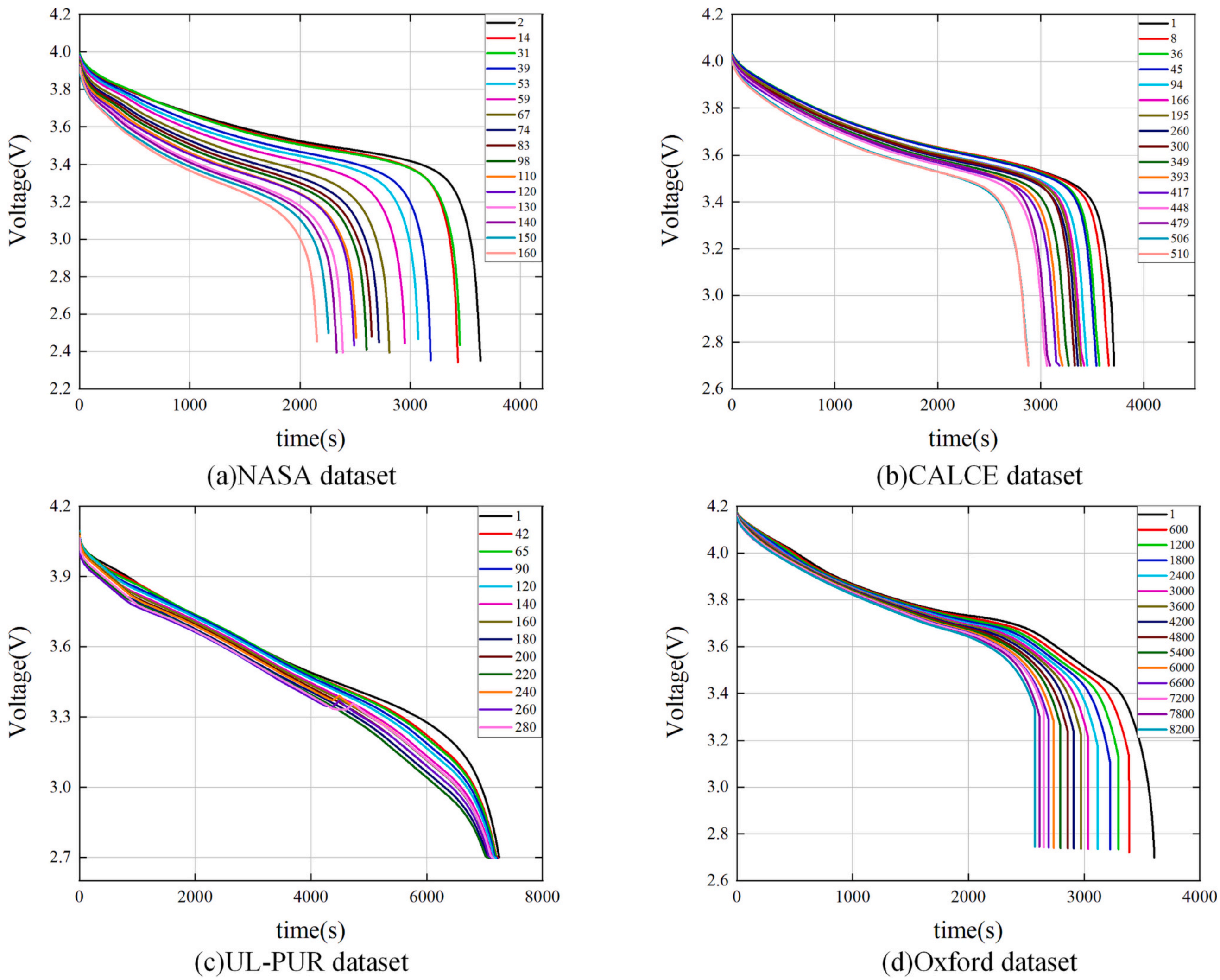


Fig. 3. The variation of voltage profiles of different datasets in CC discharging with aging cycles.

Table 1
The correlation coefficient between HFs and SOH.

Dataset	Battery serial no.	S_{ch}-SOH	S_{dis}-SOH
NASA	B0005	0.94268	0.99993
	B0006	0.94518	0.99973
	B0007	0.92037	0.90378
	B0018	0.81248	0.99976
CALCE	CS_2_35	0.90290	0.98658
	CS_2_36	0.93370	0.98981
	CS_2_37	0.93920	0.98850
UL_PUR	#1	0.91103	0.84726
	#2	0.90282	0.82478
Oxford	1_1	0.99947	0.99994
	1_2	0.98677	0.99968
	1_3	0.99948	0.99996
	1_4	0.99981	0.99996
	1_5	0.99960	0.99966
	1_6	0.99961	0.99995
	1_7	0.99973	0.99994
	1_8	0.99970	0.99997

Gaussian process prior. Both noise and Gaussian process priors consist in the GPR model, which is solved according to Bayesian inference.

A typical GPR is used to approximate the target output $f(x)$, where x is a d-dimensional output vector and the output function $f(x)$ is a probability distribution.

$$f(x) \sim GP\left(m(x), k_f\left(x, x^{'}\right)\right) \tag{4}$$

$$m(x) = ax + b \tag{5}$$

$$k_f(x, x^{'}) = \delta^2 exp\left(-\frac{(x - x^{'})^2}{2l^2}\right) \tag{6}$$

where $m(x)$ and k_f (x, x') are the mean squared error function and the covariance function respectively. δ is the hyper-parameter.

4.2. Long and short-term memory neural network

To solve the long-term dependency problem that exists with the Recurrent Neural Network (RNN) in general, a temporal recurrent neural network LSTM has been specifically designed.

The LSTM uses two gates to control the content of cell state C. One is the Forget Gate which determines the preserved amount of the cell state from the previous moment $C_{t\text{-}1}$ to the current moment C_t; the other is the Input Gate which determines the preserved percentage of the input x_t from the current network moment to the cell state C_t. Finally, Output Gate controls the output percentage of the cell state C_t to the current output value h_t of the LSTM.

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{7}$$

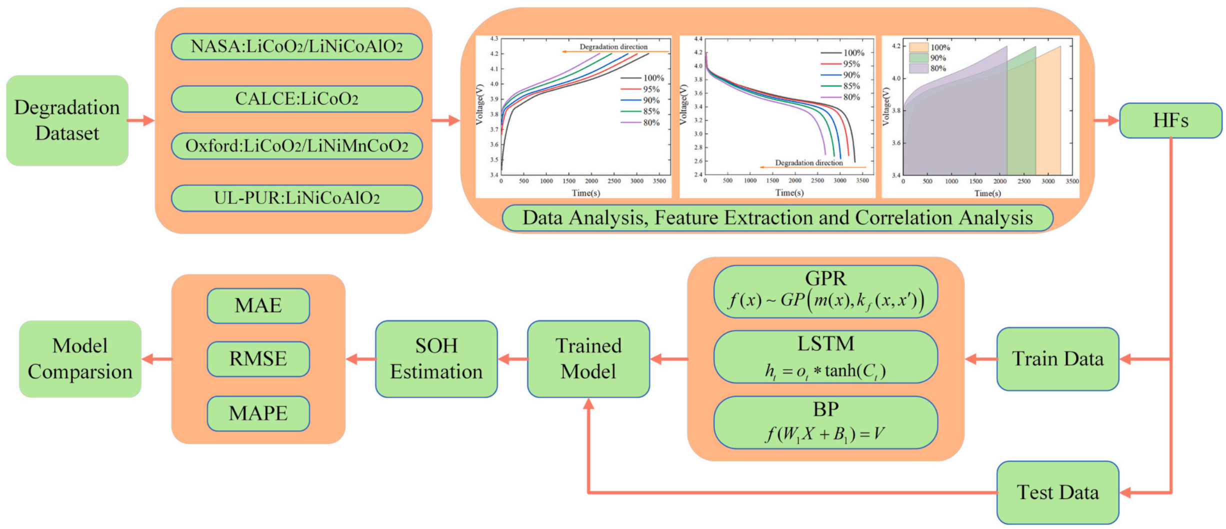


Fig. 4. The validation framework of the proposed SOH estimation model.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{8}$$

$$\widetilde{C}_t = tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{9}$$

$$C_t = f_t * C_{t-1} + i_t * \widetilde{C}_t \tag{10}$$

$$o_t = \sigma(W_o [h_{t-1}, x_t] + b_o) \tag{11}$$

$$h_t = o_t * tanh(C_t) \tag{12}$$

In the above equation, f_t is the Forget Gate, W_f is the weight matrix of the Forget Gate, $[h_{t-1}, x_t]$ denotes joining two vectors of previous output value h_{t-1} and current input x_t into a longer vector, b_f is the bias term of the Forget Gate, and σ is the sigmoid function, i_t is the Input Gate, W_i is the weight matrix of the Input Gate, and b_i is the bias term of the Input Gate. C_t is the current input cell state, tanh denotes the activation function, C_t is the cell state at the current moment and C_{t-1} is the previous cell state, and o_t represents the Output Gate.

4.3. Back propagation neural network

The BP neural network is a multi-layer feed-forward neural network, in which the signal propagates forward while the error propagates backward. The network consists of three main components, namely the input layer, the hidden layer, and the output layer.

(1) Forward Propagation. The matrix form for forward propagation is:

$$f(W_1 X + B_1) = V \tag{13}$$

$$W_2 V + B_2 = Y \tag{14}$$

where W_1 is the weight matrix from the input layer to the hidden layer, B_1 and B_2 are the threshold matrices, W_2 is the weight matrix from the hidden layer to the output layer, f is the activation function of the hidden layer, the input vector is X and the output vector is Y.

(2) Back Propagation. Define the objective function as:

$$E = \frac{1}{2} \sum_{s=1}^{M} \sum_{k=1}^{n} \left(y_k^s - o_k^s\right)^2 \tag{15}$$

where M is the number of training samples, n is the number of output layer nodes, o_k^s is the expected output of the kth output node of samples, and y_k^s is the output of the kth output node under the action of samples.

5. Experimental results and analysis

To demonstrate the effectiveness of the proposed HFs and algorithms, we used the NASA dataset, the CALCE dataset, the UL-PUR dataset, and the Oxford dataset with different types of LIBs for experimental validation.

In this experiment, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), as shown in Eqs. (16)–(18), were used to quantitatively assess the estimation performance of these two modes.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |h_i - h_i| \times 100\% \tag{16}$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (h_i - h_i)^2} \times 100\% \tag{17}$$

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{h_i - h_i}{h_i} \right| \times 100\% \tag{18}$$

where N represents the number of cycles, h_i represents the true value of SOH, and h_i represents the predicted value of SOH.

5.1. NASA dataset validation

Fig. 5(a) and (b) shows the results with its errors of SOH estimation obtained from the datasets B0005 and B0018 trained in mode 1, respectively. Table 2 shows the statistical standard results of SOH estimation for the NASA battery dataset trained in Mode 1 for all four batteries.

From Fig. 5(a) and (d), the SOH estimation error of LIBs is getting larger as aging cycles increases, while the maximum estimation error appears in LSTM. Nevertheless, the decline trend of estimated SOH is consistent with that of the real SOH. From details in Table 2, the MAE of SOH estimation in Mode 1 shows a tendency of GPR < BP < LSTM for all four batteries, while the maximum MAE of the four batteries by GPR is not more than 0.4403 %. Therefore, the proposed method can predict the SOH of LIBs with $LiCoO_2/LiNiCoAlO_2$ as the cathode material very well under Mode 1.

Fig. 5(b), (c), (e) and (f) shows the SOH estimation results with its errors of batteries B0005 and B0018 trained in Mode 2, respectively.

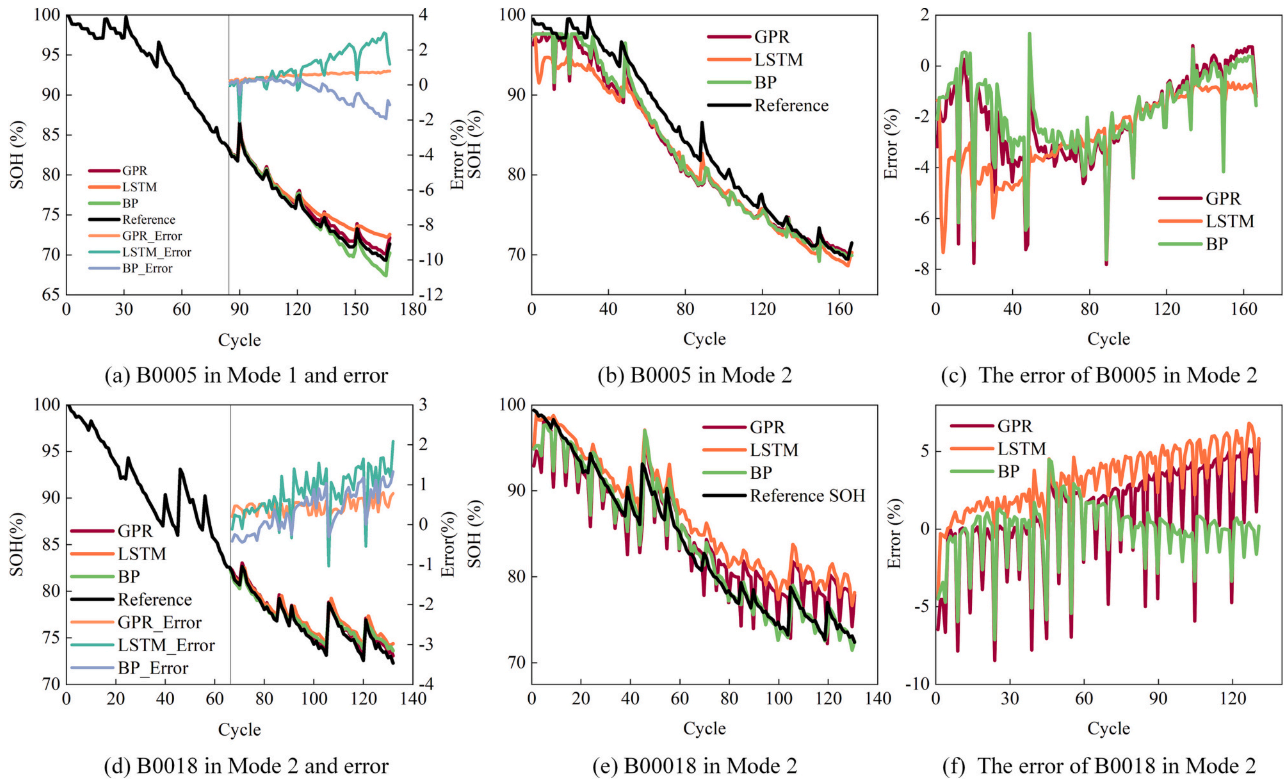


Fig. 5. The estimated SOH with errors of B0005 and B0018.

Table 2
MAE, RMSE, and MAPE of SOH estimation errors under Mode 1(%).

Battery		B0005	B0006	B0007	B0018
MAE	GPR	0.1896	0.2969	0.3992	0.4403
	LSTM	1.1921	0.8271	1.2220	0.8713
	BP	0.5311	0.4464	0.4317	0.5053
RMSE	GPR	0.2148	0.4129	0.7400	0.4761
	LSTM	1.4630	1.0505	1.4754	0.9894
	BP	0.7322	0.5993	0.7637	0.6151
MAPE	GPR	0.2394	0.4812	0.4909	0.5791
	LSTM	1.6372	1.3325	1.5634	1.1564
	BP	0.7421	0.7127	0.5433	0.6664

Similar with Mode 1, the estimated SOH variation tendency is highly consistent with the real SOH (see Table 3).

The validation experiment, it implies that the method proposed in this paper can accurately predict the SOH of LIBs with $LiCoO_2$/$LiNiCoAlO_2$ as the cathode material.

5.2. CACLE dataset validation

Fig. 6(a) shows the SOH estimation results and their errors by training CS_2_35 in Mode 1. Table 4 shows that, by GPR, LSTM and BP algorithms, the MAE of all three batteries under Mode 1 are at the same level and lower than 0.666 %, which infers the proposed method can accurately estimate the SOH of the LIBs with $LiCoO_2$ cathode under Mode 1.

Fig. 6(b) and (c) shows the estimated SOH and its errors of CS_2_35 in Mode 2. Similar to the NASA dataset, the estimated trajectory is basically consistent with the real SOH degradation profile. Together with the low errors level in Tables 4 and 5, it shows the effectiveness of proposed method for SOH estimation of LIBs with $LiCoO_2$ cathode.

Table 3
MAE, RMSE, and MAPE of SOH estimation errors under Mode 2(%).

Battery		B0005	B0006	B0007	B0018
MAE	GPR	2.2129	2.6704	2.8827	2.7049
	LSTM	2.7212	4.8119	3.1765	3.2656
	BP	1.9145	1.7267	2.6065	1.3787
RMSE	GPR	2.7458	3.7887	3.4535	3.3688
	LSTM	3.1102	5.0543	3.3646	3.7199
	BP	2.4099	2.2909	2.8198	1.9955
MAPE	GPR	2.5303	3.2670	3.9641	3.3281
	LSTM	3.0646	6.4023	3.8193	4.1065
	BP	2.2929	2.2188	2.9604	1.6073

5.3. UL-PUR dataset validation

Fig. 6(d) shows the estimated SOH and errors by battery #2 in Mode 1. As the aging cycles increase, the errors between the estimated SOH and the true value become increasingly larger, but the error still stays within a certain small range. From Table 6, both ternary LIBs with the $LiNiCoAlO_2$ cathode can be accurately estimated in Mode 1.

Fig. 6(e) and (f) shows the SOH estimation results of #2 in Mode 2 with the true SOH values. It is founded, GPR has an increasing tendency to error with the aging circles, while the LSTM and BP has a decreasing error. In Table 7, the maximum MAE #2 trained with Mode 2 does not exceed 0.4662 %. The results show that the estimation model can accurately estimate SOH of LIBs.

5.4. Oxford dataset validation

Fig. 7(a) and (d) shows the SOH estimation results and errors of batteries 1_3 and 1_8 under Mode 1 by GPR, LSTM and BP, respectively. Table 8 shows the SOH estimation errors of all eight batteries trained with Mode 1.

From Fig. 7(a) and (d), the results obtained for batteries 1_3 and 1_8 trained under Mode 1 match very well with the real SOH decline tendency. From Table 8, the MAE for all batteries is smaller than 0.7442 % under the different algorithms. It proves that the proposed method can

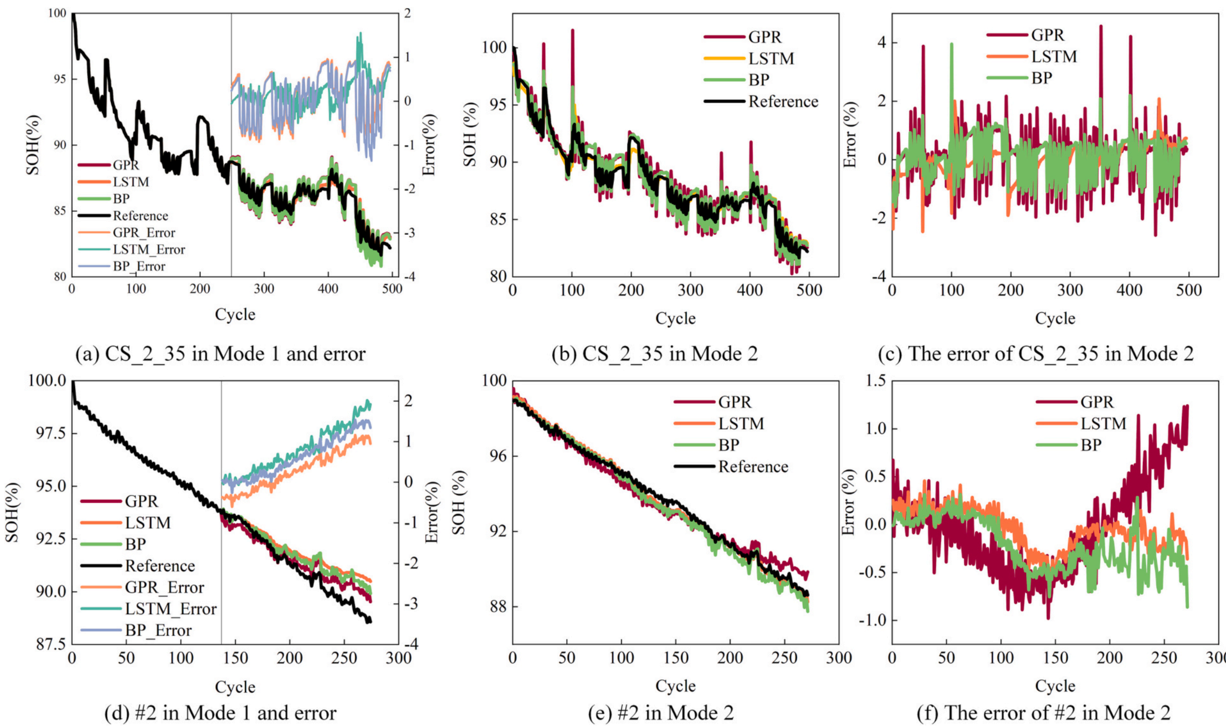


Fig. 6. The estimated SOH with errors of CS_2_35 and #2.

Table 4
MAE, RMSE, and MAPE of SOH estimation errors under Mode 1(%).

Battery		CS_2_35	CS_2_36	CS_2_37
MAE	GPR	0.4999	0.5290	0.5220
	LSTM	0.2710	0.6436	0.6660
	BP	0.5118	0.5278	0.5779
RMSE	GPR	0.5657	0.6548	0.6538
	LSTM	0.3732	0.6548	0.8694
	BP	0.5838	0.5968	0.6985
MAPE	GPR	0.5842	0.6115	0.6158
	LSTM	0.3198	0.7613	0.7934
	BP	0.6003	0.6112	0.6794

Table 5
MAE, RMSE, and MAPE of SOH estimation errors under Mode 2(%).

Battery		CS_2_35	CS_2_36	CS_2_37
MAE	GPR	0.6837	0.9161	0.6273
	LSTM	0.5573	0.3240	0.5474
	BP	0.5614	0.8217	0.5414
RMSE	GPR	0.9518	1.0969	0.7789
	LSTM	0.7186	0.4949	1.4961
	BP	0.6804	1.0004	0.6565
MAPE	GPR	0.7749	1.0280	0.7146
	LSTM	0.6461	0.3623	0.6227
	BP	0.5432	0.9161	0.6474

Table 6
MAE, RMSE, and MAPE of SOH estimation errors under Mode 1(%).

Battery		#1	#2
MAE	GPR	0.7950	0.4664
	LSTM	0.7806	0.7921
	BP	0.3514	0.6188
RMSE	GPR	1.0031	0.5673
	LSTM	1.0561	0.9759
	BP	0.4964	0.7902
MAPE	GPR	0.8910	0.5160
	LSTM	0.8769	0.8789
	BP	0.3915	0.6796

Table 7
MAE, RMSE, and MAPE of SOH estimation errors under Mode 2(%).

Battery		#1	#2
MAE	GPR	0.4106	0.3842
	LSTM	0.2132	0.1744
	BP	0.2500	0.2822
RMSE	GPR	0.5647	0.4662
	LSTM	0.2413	0.2102
	BP	0.3263	0.3442
MAPE	GPR	0.4451	0.4149
	LSTM	0.2278	0.1859
	BP	0.2714	0.3068

estimate the SOH of LIBs with $LiCoO_2$/$LiNiMnCoO_2$ cathode very well under Mode 1.

Fig. 7(b), (c), (e) and (f) shows the SOH estimation results and errors for batteries 1_3 and 1_8 under Mode 2, while the SOH estimation errors of all eight batteries are displays in Table 9.

Tables 8 and 9 demonstrate the effectiveness of the proposed method by HF extracted from the CC charge/discharge curve under GPR, LSTM, and BP algorithms. It validates the method for LIBs with $LiCoO_2$/$LiNiMnCoO_2$ as the cathode.

Through the above experimental results, it can conclude that the HFs proposed in this paper do not only merely correlate very strongly with the discharge capacity, but also obtain great estimation results regardless of the input algorithm. In addition, this model is generally applicable to different types of LIBs, which is also verified in the above four battery datasets. This method only requires real-time monitoring of the voltage, and current during battery charging and discharging, so only simple sensors are required instead of complex monitoring units. On the other hand, the excellent generalization ability of proposed model was validated using cross-validation method.

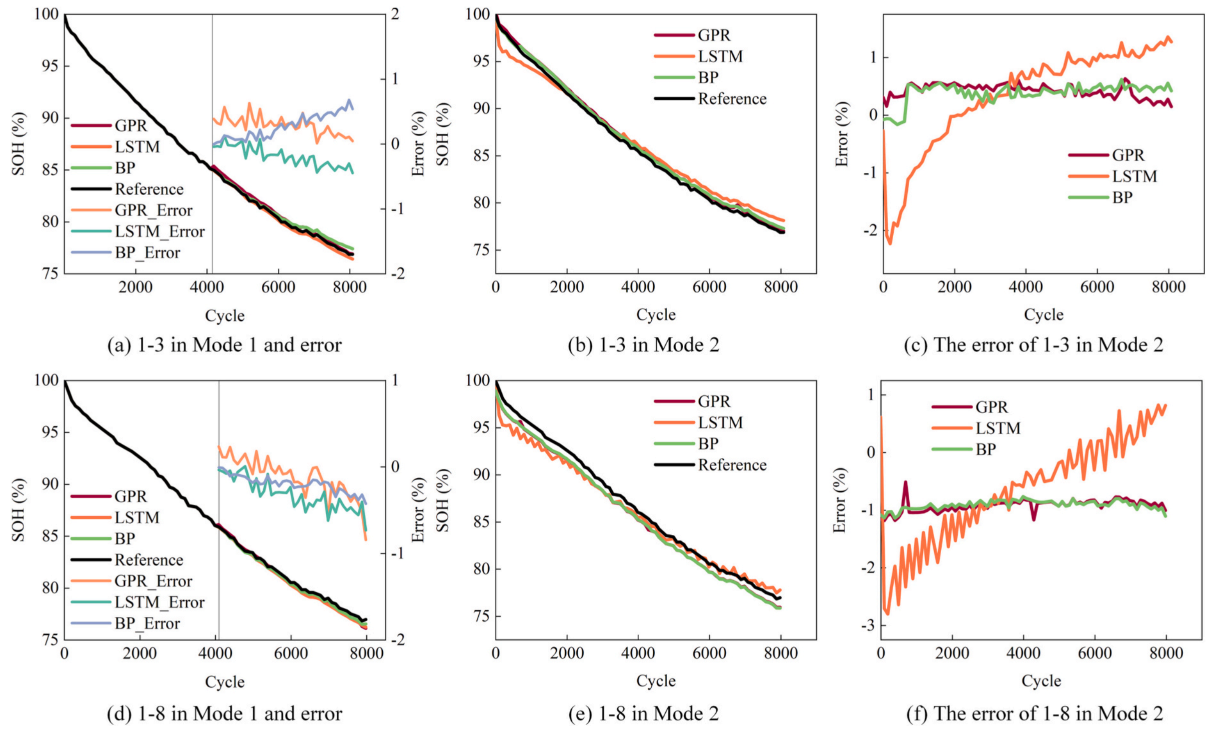


Fig. 7. The estimated SOH with errors of 1_3 and 1_8.

Table 8
MAE, RMSE, and MAPE of SOH estimation errors under Mode 1(%).

Battery	MAE			RMSE			MAPE		
	GPR	LSTM	BP	GRP	LSTM	BP	GPR	LSTM	BP
1_1	0.3456	0.5529	0.4357	0.3681	0.6209	0.4736	0.4348	0.7096	0.5617
1_2	0.7442	0.6141	0.2577	0.8320	0.7892	0.3020	0.9450	0.7894	0.3328
1_3	0.2784	0.2201	0.2753	0.3104	0.2551	0.3310	0.3443	0.2785	0.3480
1_4	0.3194	0.1293	0.0815	0.4178	0.1573	0.0976	0.3880	0.1581	0.0991
1_5	0.3586	0.1086	0.1619	0.3723	0.1346	0.1932	0.4223	0.1282	0.1906
1_6	0.2162	0.3507	0.2989	0.3552	0.5167	0.4327	0.2587	0.4271	0.3588
1_7	0.1835	0.3145	0.2940	0.2195	0.3765	0.3322	0.2218	0.3833	0.3560
1_8	0.1981	0.3344	0.2040	0.2664	0.3783	0.2252	0.2498	0.4209	0.2572

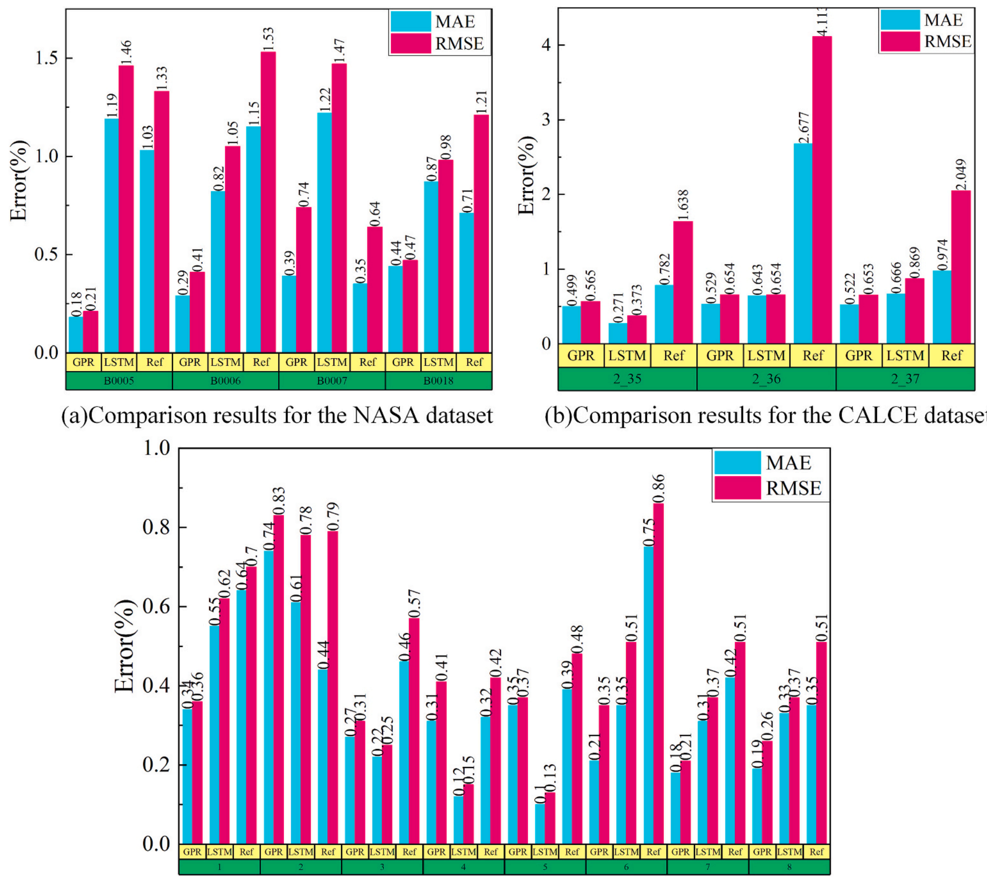
Table 9
MAE, RMSE, and MAPE of SOH estimation errors under Mode 2(%).

Battery	MAE			RMSE			MAPE		
	GPR	LSTM	BP	GRP	LSTM	BP	GPR	LSTM	BP
1_1	1.0683	1.0358	0.8707	1.0733	1.0736	0.8859	1.2709	1.2530	1.0223
1_2	0.5931	0.9614	0.4045	0.5268	1.3604	0.5781	0.4493	1.2075	0.5077
1_3	0.4106	0.8317	0.3988	0.4268	0.9627	0.4175	0.4761	0.9796	0.4697
1_4	0.2202	0.5481	0.3684	0.2318	0.6294	0.3926	0.2543	0.6435	0.3684
1_5	0.4748	0.6281	0.4344	0.4847	0.7762	0.4519	0.5331	0.6281	0.4879
1_6	0.1300	0.8701	0.1701	0.1418	1.0773	0.2013	0.1462	0.0947	0.1932
1_7	0.9215	0.5879	0.8704	0.9215	0.6808	0.8737	1.0194	0.6481	0.9987
1_8	0.9381	0.8809	0.9204	0.9386	1.1125	0.9238	1.0772	0.9738	1.0785

5.5. Models comparison

To further validate the accuracy of the SOH estimation method in this paper, we compared the estimation results with reference data from other literatures under Mode 1 training. For well comparison, we chose the SOH estimation results of the GPR algorithm and another algorithm with the worst accuracy in above sections to compare with that in the public literature under the same conditions. Here, the NASA dataset, the CALCE dataset and the Oxford dataset were selected for algorithm comparison, as there are no papers or reports available on SOH estimation by the UL-PUR dataset till now.

First for the NASA dataset, we compared the SOH estimation errors of the GPR, which has the best accuracy, and the LSTM with the worst accuracy in Section 5.1, with the reference [26]. Fig. 8(a) shows the estimation results of the GPR are much better than those of the reference, while the estimation errors of the LSTM and the reference are at the same level. The comparison indicates, the SOH estimation based on the area of the charge-discharge curve proposed in this paper, has a



(a)Comparison results for the NASA dataset

(b)Comparison results for the CALCE dataset

(c)Comparison results for the Oxford dataset

Fig. 8. Comparison results for the different datasets.

better SOH estimation for the four batteries in the NASA dataset compared to the reference.

For the CALCE dataset, it can be obtained that the estimation errors of the GPR with smaller errors and the LSTM with larger errors in Section 5.2 are both much lower than the results in reference [28] as shown in Fig. 8(b). This proves that the proposed method has a higher estimation accuracy for SOH in the CALCE dataset compared to reference.

From Tables 8 and 9, the GPR, BP, and LSTM algorithms have essentially the similar errors and for Oxford dataset. Therefore, Fig. 8(c) compares the error of GPR and LSTM with that of reference data [34]. It is found that the error of both GPR and LSTM is smaller than the error in reference. Especially, the MAE of battery 1_6 in reference is twice the MAE of LSTM in Table 8. Therefore, the proposed method has better results for the SOH estimation of the eight batteries in the Oxford dataset compared to the reference Through the algorithms comparison, by only two HFs, namely the areas of the charge and discharge voltage curve respectively, the SOH estimation has a better applicability and a higher accuracy for the three commonly used algorithms.

6. Conclusion

To improve the accuracy and reduce the computational complexity of real-time SOH estimation of LIBs, this paper proposed a data-driven SOH estimation method based on the area of the CC charging and discharging voltage curves of LIBs as HFs, and validated the method under four typical battery datasets, and then selected HFs of the battery and its SOH shown a strong correlation, which was all greater than 0.81. Then the two HFs were used as the input of the GPR, LSTM, and BP algorithms respectively for SOH estimation. The SOH estimation results demonstrated that the BP neural network had the greatest estimation accuracy overall, regardless of whether it was trained under mode 1 or mode 2. Moreover, the two other algorithms also show good accuracy, which indicates that the two selected HFs have a good mapping relationship to SOH of LIBs. Meanwhile, by comparing with other methods under three datasets, the proposed estimation method has a higher accuracy with only two HFs, which can significantly improve the real-time estimation performance.

CRediT authorship contribution statement

Jiang Wu: Writing - Review & Editing, Software. **Zelong Liu:** Data curation, Methodology, Writing - Original draft preparation. **Yan Zhang:** Visualization. **Dong Lei:** Data Curation. **Bo Zhang:** Validation. **Wen Cao:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This paper used open datasets.

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